

Cross validation:

- ↳ It is a resampling method that uses different portions of the data to test and train a model on different iterations.
- ↳ The goal of cross-validation is to test the model's ability to predict new data that was not used in estimating it.
- ↳ In order to flag problem like overfitting or selection bias, use it.
- ↳ It gives an insight on how the model will generalize to an independent dataset (i.e. an unknown data set, for instance from a real problem).
- ↳ In machine learning, we couldn't fit the model on the training data and can't say that the model will work ~~so~~ accurately for real data.
- ↳ For this, we must assure that our model got the correct patterns from the data.
- ↳ For this purpose, we use the cross-validation technique.

→ cross validation is a technique used in machine learning to evaluate the performance of a model on unseen data.

↳ It involves dividing the available data into multiple k folds or subsets, using one of these folds as a validation set and training the model on the remaining folds.

↳ This process is repeated multiple times, each time using a different fold as the validation set.

Type of cross-validation

- k -fold cross validation
- Leave-one-out cross, (LOOCV)
- Holdout
- Stratified cross

1) Hold out validation

pros	→ simple, easy to understand & implement
cons	→ Not suitable for imbalanced dataset

↳ 50% of given dataset for training

↳ 50% of test " testing

↳ higher bias (drawback)

↳ left some important information / pattern during training.

How do the cross validation.

Step of k -fold cross-validation

↳ split the i/p dataset into k groups

↳ For each group.

↳ Take one group as the ~~test~~ reserve or test data set.

↳ Use remaining groups as the training dataset

↳ Fit the model on the training set and evaluate the performance of the model using the test set

→ Let's take an example of 5-fold cross validation. So, the dataset is grouped into 5 fold.

→ On 1st iteration, the first fold is reserved for test the model, and rest are used to train the model. On 2nd iteration

→ On 2nd iteration the second fold is used to test the model and rest are used to train the model.

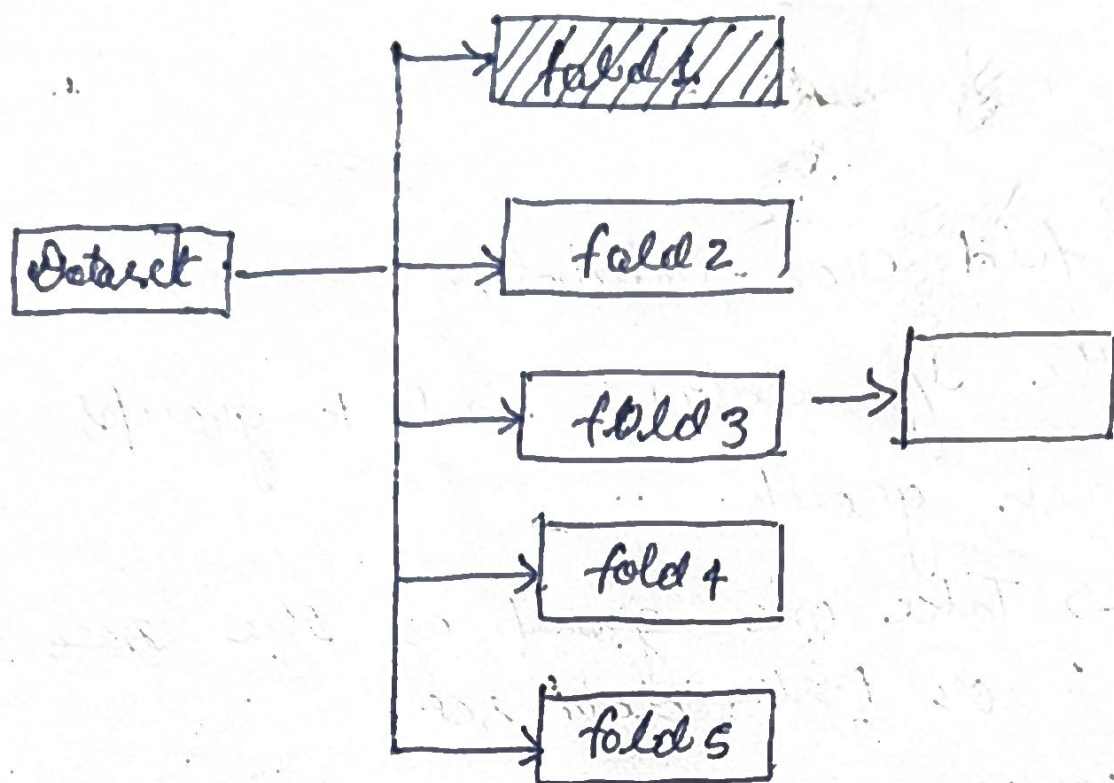
→ This process will continue until each fold is not used for the test fold.

Pros: ① The model has low bias.

② low time complexity.

③ Entire dataset use for both training & testing.

Cons: ① Not suitable for imbalance dataset.



LOOC (Leave One Out Cross Validation)

↳ Model is trained on entire dataset except for one data point which is used for testing.

↳ This process (above process) is repeated for each data point in the dataset.

Thus, All data points are used for training, resulting in low bias.

↳ Testing on a single data point can cause high variance, especially if the point is outlier.

↳ It can be very time consuming for large datasets, as it requires one iteration per data point.

Pro: Simple, easy to understand & implement.

Cons: Computation time require is high.

Stratified K-fold

- ↳ This technique is similar to K-fold cross validation with some little changes.
- ↳ This approach works on stratification concept, it is process of rearranging the data to ensure that each fold or group is good representative of the complete dataset.
- ↳ To deal with the bias and variance, it is one of the best approaches.
- ↳ It can be understood with an example of housing prices, such that the price of some houses can be much high than other houses.
- ↳ To tackle such situations, a stratified K-fold cross-validation techniques is useful.
- ↳ Each fold of this process (stratified K-fold) has same class distribution as the full dataset.
- ↳ This is useful for imbalanced datasets where some classes are underrepresented.
- ↳ The dataset is divided into K-fold, keep class proportions consistent in each fold.
- ↳ In each iteration, one fold is used for testing and remaining fold for training.

↳ This process is repeated k times so that each fold is used once as test set.

↳ help classification models generalize better by maintaining balanced class representation

Pros: Work well for imbalance dataset

Cons: Not suitable for timeseries